

roadway intersections.^{40,41} CAL3QHC combines the CALINE model with a traffic algorithm to estimate delays and queues that occur at signalized intersections. This model is used as a screening model and is the EPA-recommended model for this application.

EPA also developed the CAL3QHCR model, which is a more refined version of CAL3QHC that requires local hourly meteorological data.⁴² CAL3QHCR incorporates enhancements to process up to a year of meteorological data and emissions data that vary by day of the week and hour of day. The model has the capability, with the use of various concentration-averaging algorithms, to predict 1-hour, 8-hour, 24-hour, and annual concentrations, compared with only the maximum hourly average computed by CAL3QHC. CAL3QHCR is the EPA-recommended model for PM (i.e., PM₁₀ and PM_{2.5}) hot-spot modeling for highway and roadway intersection projects/action and also EPA-recommended for CO when a CAL3QHC analysis with conservative assumptions indicates a potential to exceed the NAAQS.

For further information on CAL3QHC/CAL3QHCR model inputs and guidance on how to conduct a CO and PM hot-spot analysis see **Section 7.2** (*Roadway Dispersion Modeling*) and **Appendix E** (*Roadway Dispersion Modeling [Hot-Spot Analysis]*).

⁴⁰ EPA Preferred/Recommended Models, *CALINE3*,
http://www.epa.gov/ttn/scram/dispersion_prefrec.htm#caline3.

⁴¹ EPA, *User's Guide to CAL3QHC Version 2.0: A Modeling Methodology for Predicting Pollutant Concentrations Near Roadway Intersections*, November 1992 [EPA-454-R-92-006],
<http://www.epa.gov/ttn/scram/userg/regmod/cal3qhcug.pdf>.

⁴² EPA Preferred/Recommended Models, *CAL3QHC/CAL3QHCR*,
http://www.epa.gov/ttn/scram/dispersion_prefrec.htm#cal3qhc.